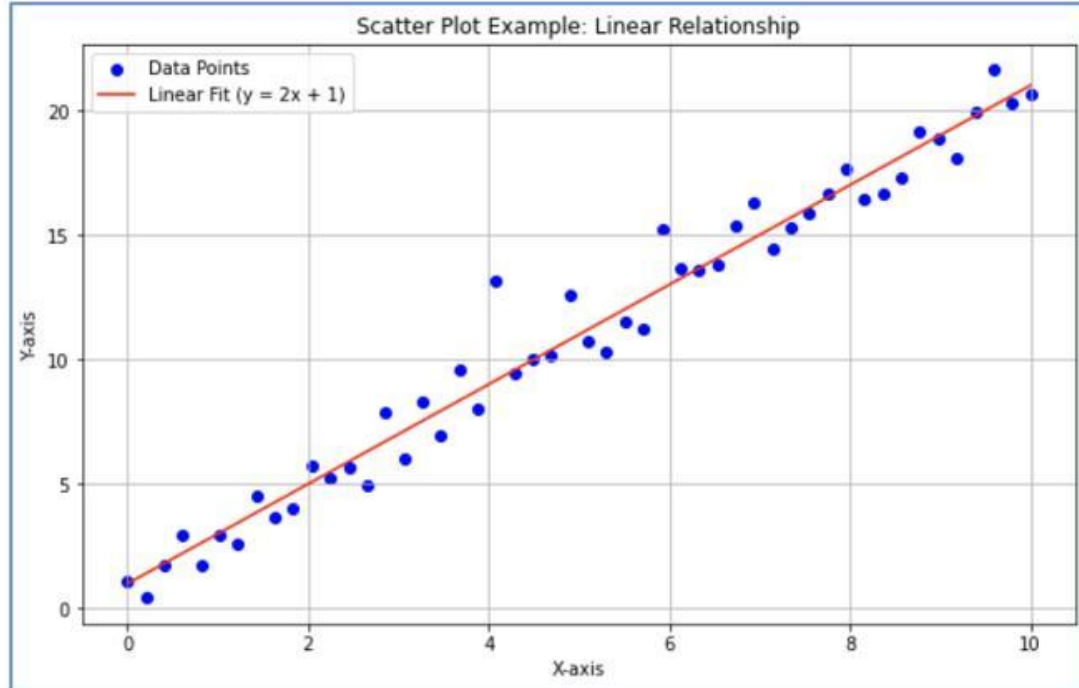


Plots in Data Analytics

Scatter Plot (Relationship Between Variables)



It shows

Relationship between two variables

Pattern: linear / non-linear

Importance

Detect **correlation** (like in your PCA example)

Identify clusters and outliers

Line Plot (Trend Over Time)



It shows

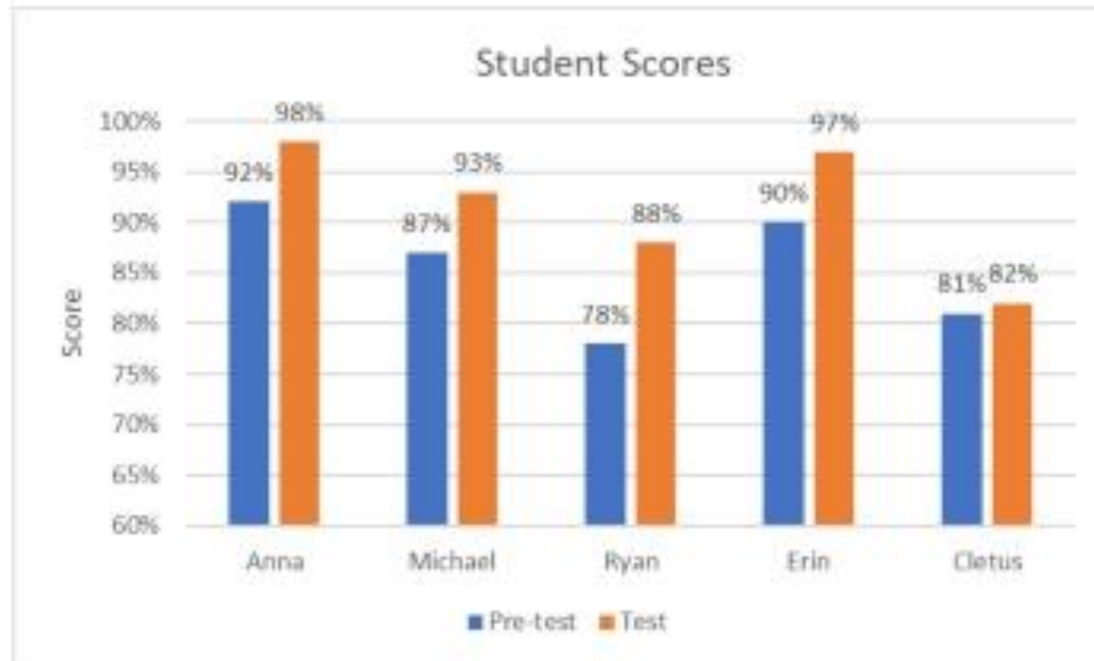
Trends over time

Importance

Used in **forecasting**

Helps detect **growth/decline patterns**

Bar Chart (Comparison Between Categories)



It shows

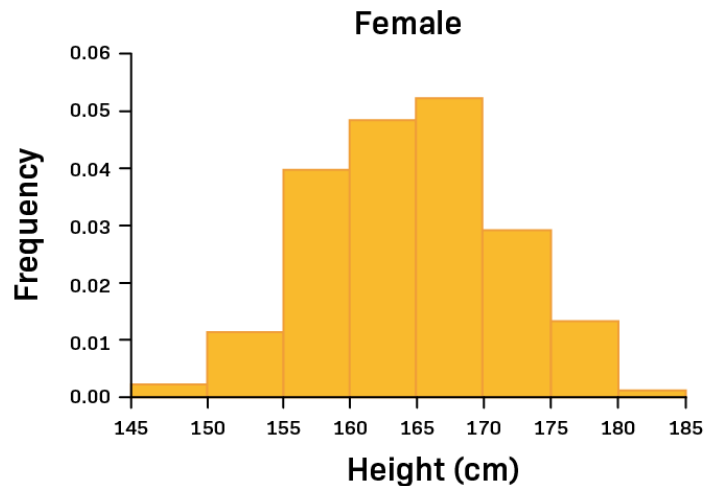
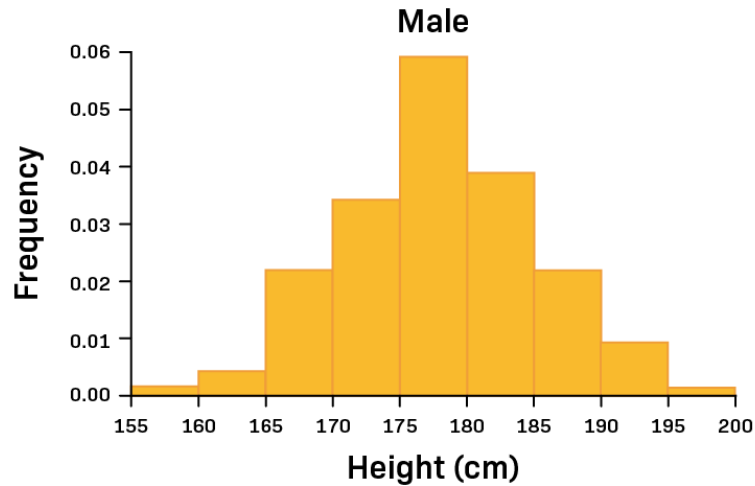
Comparison across categories

Importance

Easy to compare values

Widely used in reports

Histogram (Distribution of Data)



It shows

Frequency distribution

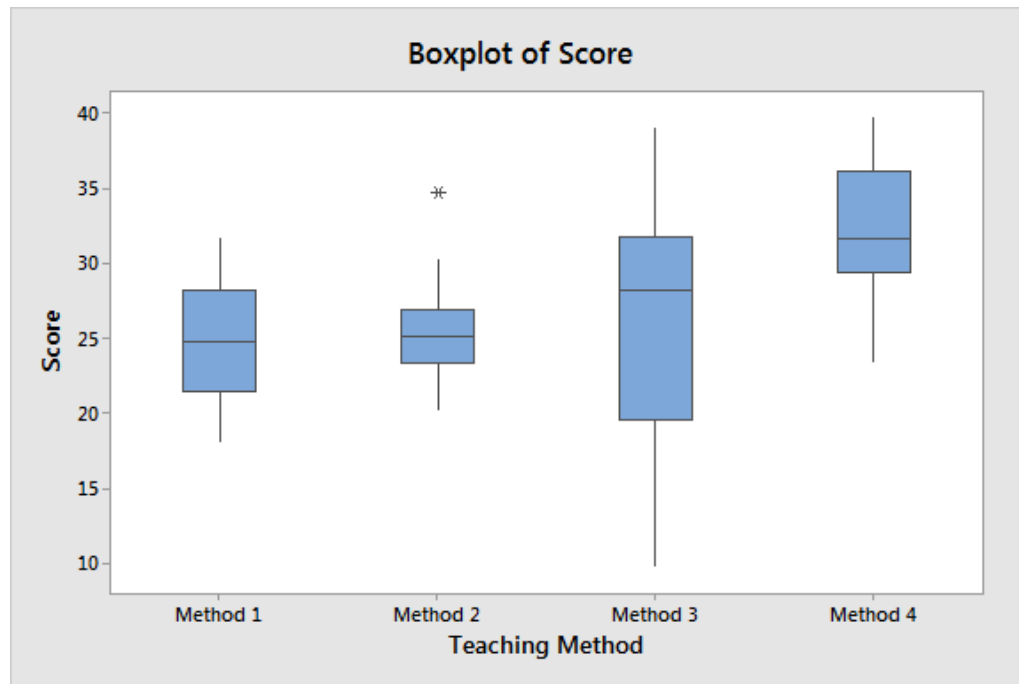
Shape of data (normal, skewed)

Importance

Understand **data spread**

Important before applying PCA/ML

Box Plot (Outliers & Spread)



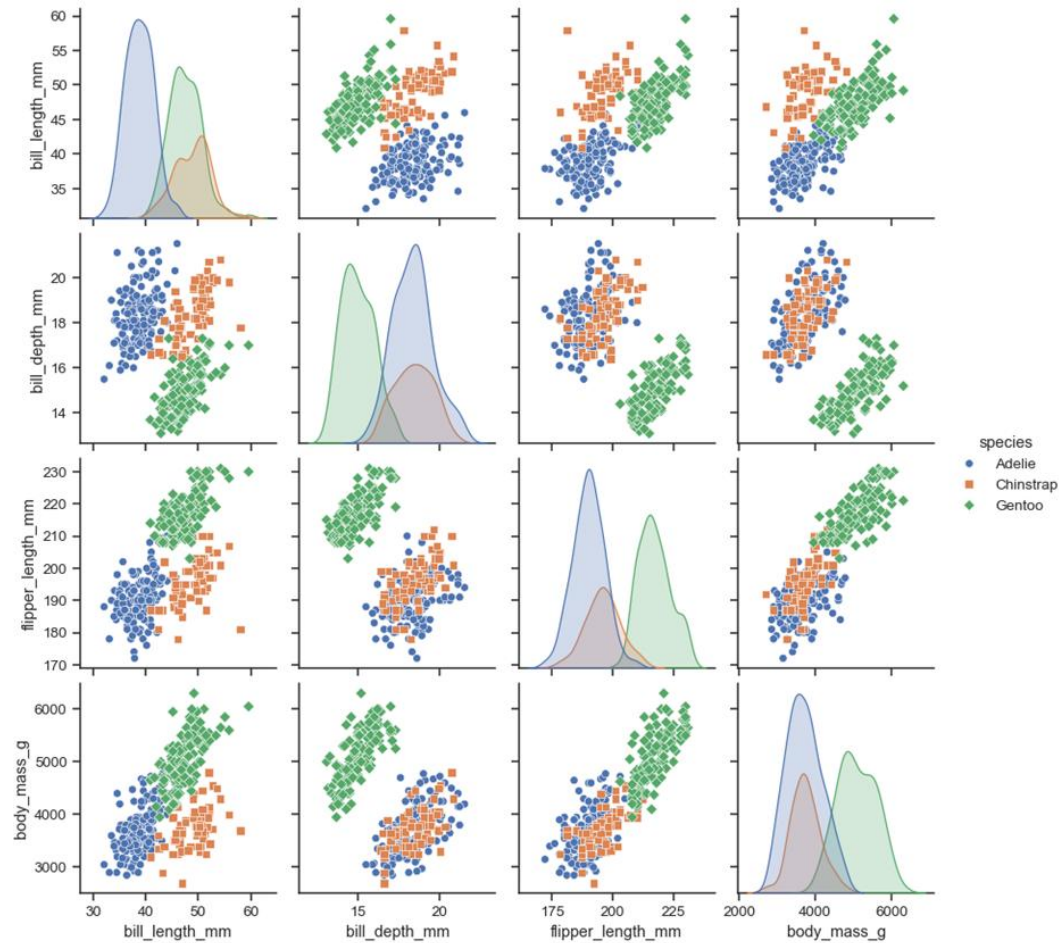
It shows

Median, quartiles
Outliers (100 here)

Importance

Detect **extreme values**
Helps in **data cleaning**

Pair Plot (Used in PCA Context)



It shows
Pairwise relationships

Importance
Detect **multicollinearity**
Helps decide if **PCA is needed**

